

Autodesk® Scaleform®

HUD

Scaleform 4.3 HUD AAA . HUD Flash UI
C++ .

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:
: 2010 7 30

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Autodesk® Scaleform® 4.3

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HUD Kit

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1 소개

Scaleform HUD(Heads Up Display) UI AAA
Scaleform HUD Scaleform® Direct Access API
(FPS) UI

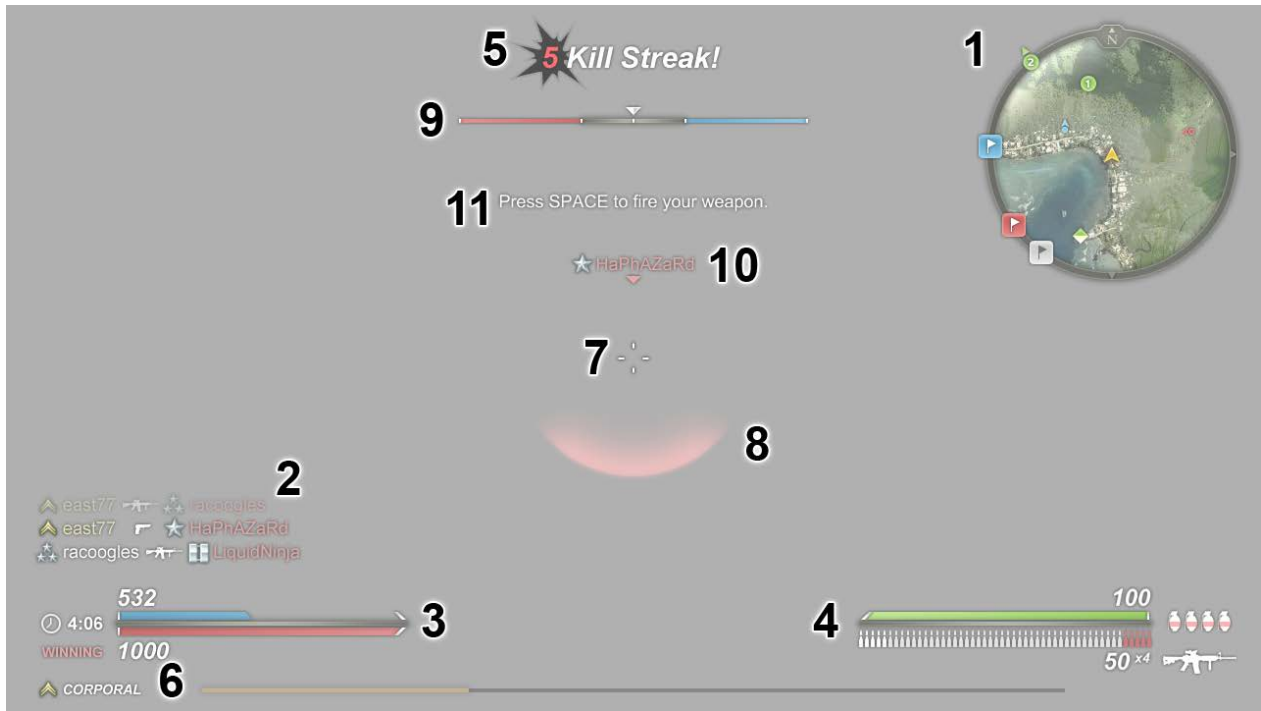


그림 1: HUD 개요

Scaleform HUD UI

- 1.
- 2.
- 3.
- 4. / /
- 5.
- 6.
- 7.
- 8.
- 9.

10. /

11.

Scaleform HUD

FPS

UI

,

2 개요

2.1

- *Apps\Kits\HUD* - HUD C++
Bin\Data\AS2 or AS3\Kits\HUD - Flash asset ActionScript (AS2
ActionScript 2/Flash 8, AS3 ActionScript3/Flash10.)
- *Projects\Win32\{Msvc80 or Msvc90}\Kits\HUD* - Visual Studio 2005/2008
Windows
- *Projects\Xbox360\{Msvc80 or Msvc90}\Kits\HUD* - Visual Studio 2005/2008
Xbox 360
- *Projects\Common\HudKit.mk* - HUD PS3 makefile.

Bin\Kits\HUD

Windows

HUDKit.exe

Windows

Projects\Win32\Msvc80\Kits

Projects\Win32\Msvc90\Kits

Scaleform 4.3 Kits.sln

"

"

Bin\Data\AS2\Kits\HUD or

Bin\Data\AS3\Kits\HUD

Xbox 360

Projects\Xbox360\{Msvc80}\Kits\HUD

Projects\Xbox360\{Msvc90}\Kits\HUD

Scaleform

4.3 Kits.sln

(asset) *Bin\Data\AS2 or*

AS3\Kits\HUD

Xbox 360

```

PS3      Scaleform      make      .
        HUD      . PS3
        SN      Bin\PS3      ,
o  app_home/ Directory: {Local Scaleform Directory}\Bin\Data\AS2 or AS3\Kits\HUD

```

2.2

HUD



2.3

2.3.1 키보드(Windows)

W, A, S, D	. W, S . A, D . .
SPACE	
R	.
G	.
ESC	.
B	AI .
1, 2	. • 1 . • 2 .

2.3.2 컨트롤러(Xbox/PS3)

방향 패드	. W, S
왼쪽 조이스틱	. A, D
오른쪽 트리거	.
X 단추 / 네모 버튼	.
왼쪽 트리거	.
B 단추 / 동그라미 버튼	.
Y 단추 / 세모 버튼	.

3 아키텍처

HUD, Flash UI asset, C++ HUD, C++

3.1 C++

C++ HUD HUD, Flash UI
 MVC, Fx HUD
 HUD HUD HUD
 HUD HUD HUD
 HUD HUD
 Scaleform Direct Access API UI
 Scaleform
 C++ *Apps\Kits\HUD*

3.1.1 데모

- **HUDKitDemo.cpp** – Scaleform Player, C++
 , SWF
- **HUDKitDemoConfig.** – FxPlayerConfig.h HUDKitDemo
 Windows

3.1.2 HUD/미니맵 뷰 코어

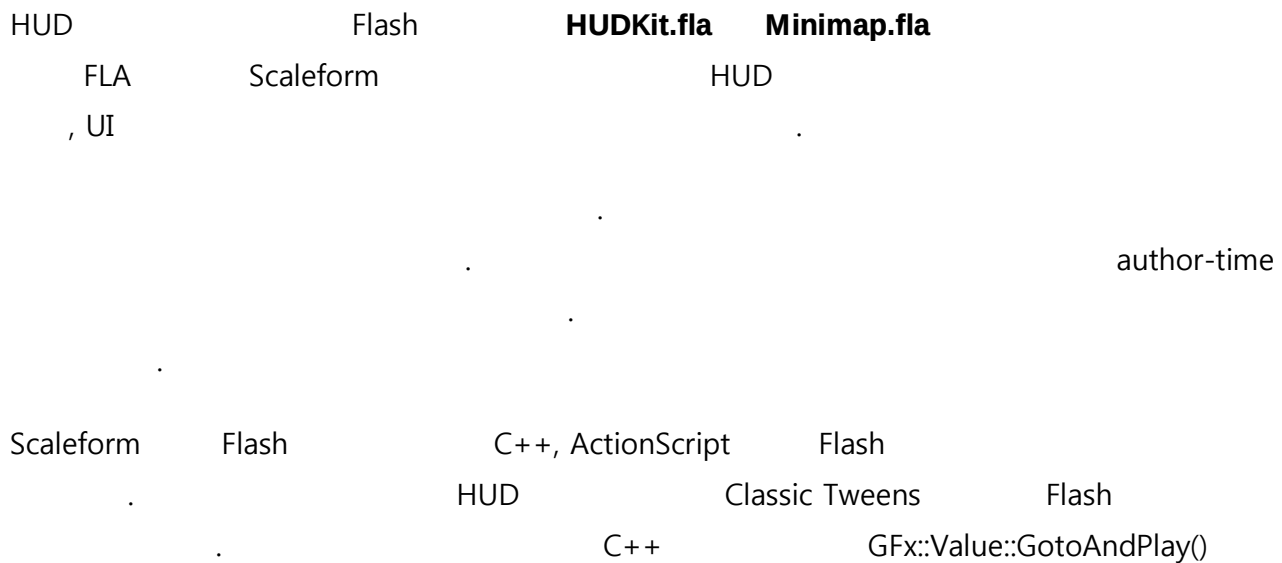
- **FxHUDKit.h** – HUD
- **FxHUDView.h/.cpp** – HUD

- **FxMinimap.h** –
- **FxMinimapView.h/.cpp** –

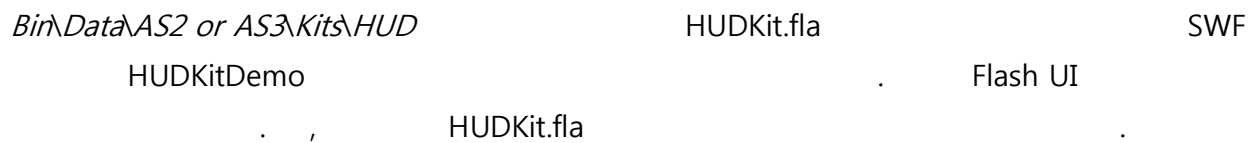
3.1.3 게임 시뮬레이션

- **HUDAdapter.h/.cpp** – HUD
- **HUDEntityController.h/.cpp** – AI
- **HUDSimulation.h/.cpp** – UI

3.2 *Flash*

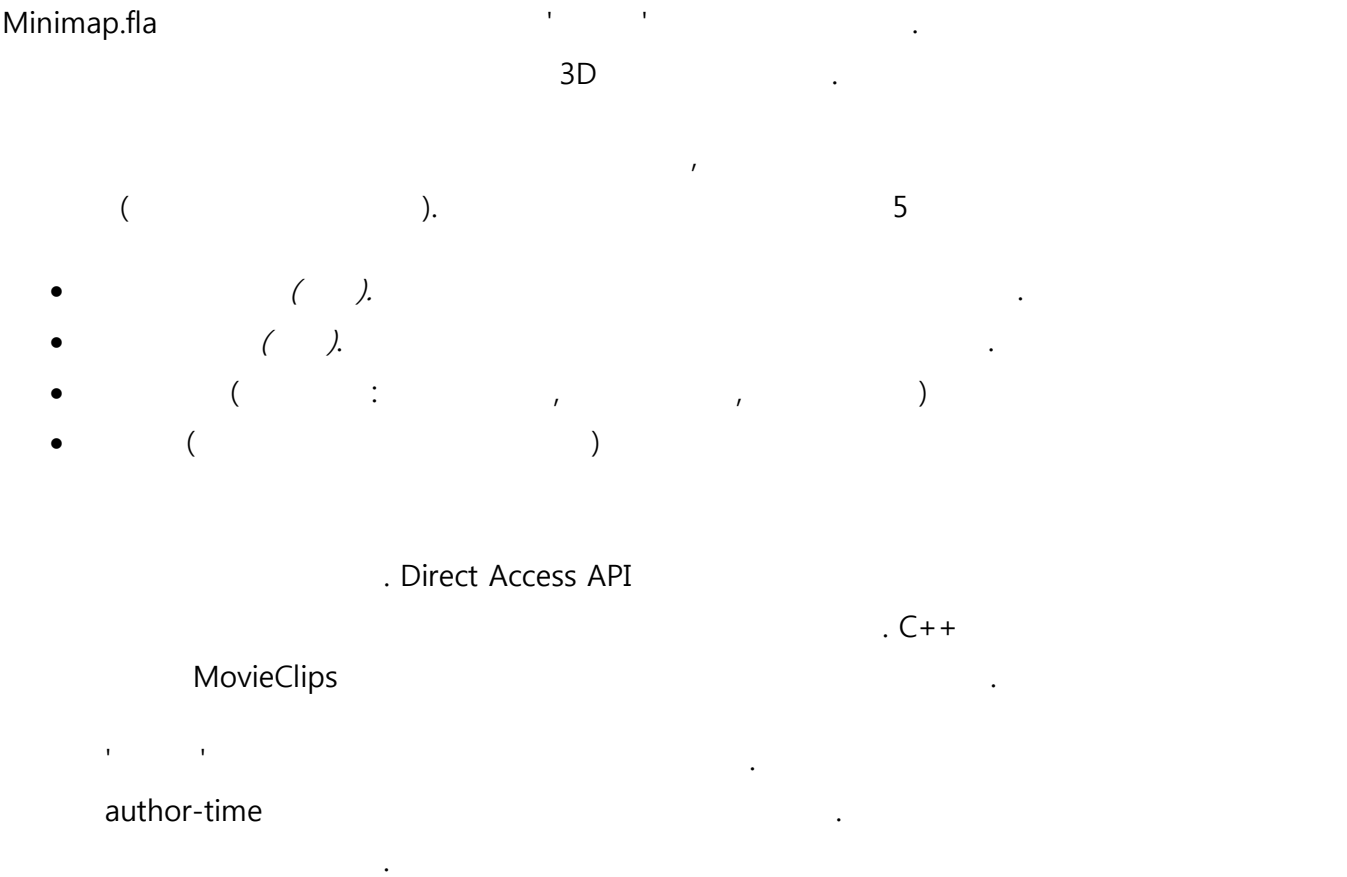


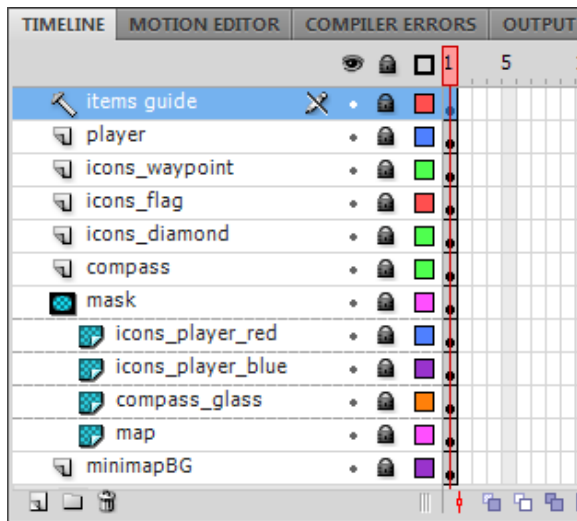
3.2.1 HUDKit.fla



3.2.2 Minimap.fla

그림 3: Minimap 심볼





그 4: '미니맵' 심볼을 구성하는 레이어

- **items guide:**
SWF
- **player:**
- **icons_waypoint:**
- **icons_flag:**
- **icons_diamond:**
- **compass:** ()
- **mask:**
 - **icons_player_red:** ()
 - **icons_player_blue:** ()
 - **compass_glass:**
 - **map:**

C++

- **minimapBG:**

C++

Bin\Data\AS2 or AS3\Kits\HUD\com\scaleform

com.scaleform.Minimap

HUDKitDemo

4 HUD 뷰

C++ HUD HUDKit.swf HUD

- FxHUDView – HUD (, HUDKit.swf) .
MovieClips
- BillboardCache – / ()
- FxHUDLog – HUD . FxHUDMessages
- FxHUDMessage – MovieClip .

FxHUDView MovieClip MC(MovieClip) .

FxHUDView UpdateView()
MovieClip . MovieClip , HUDKit fla
UI .

FxHUDView::UpdateView()
UpdateTeamStats(),
UpdatePlayerEXP(), Update PlayerStats(), UpdateEvents(), UpdateTutorial() UpdateBillboards().
UI .

UpdateView() MovieClip .
MovieClip MovieClips

Scaleform Direct Access API Flash
Gfx::Value Scaleform Flash
Gfx::Value::SetDisplayInfo(),
Gfx::Value::GetDisplayInfo() Gfx::Value::SetText() MovieClips
HUD .

4.1

```
teamStats MovieClip    HUDKit.fla    Scene 1    teamStats  
    ,  
    textFields(redWinning, blueWinning),  
textField              (scoreRed, scoreBlue),      (teamRed, teamBlue)  
    runtime clock textField(roundTime)
```

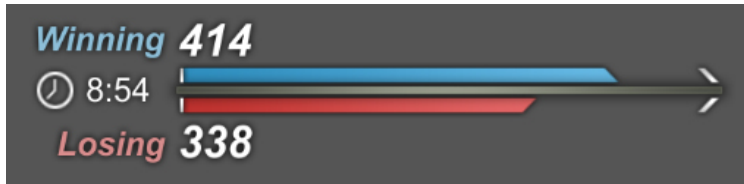


그림 5: teamStats 심볼

```
teamStats              FxHUDView::UpdateTeamStats()  
teamStats MovieClip
```

```
void FxHUDView::UpdateTeamStats(FxHUDEnvironment *penv)  
{  
    // Retrieve the scores from the simulation.  
    unsigned scoreBlue = unsigned(penv->GetHUDBlueTeamScore());  
    unsigned scoreRed = unsigned(penv->GetHUDRedTeamScore());  
  
    String text;  
    Value tf;  
    Value::DisplayInfo info;  
  
    // We won't update any element who has not changed since the last update  
    // to avoid unnecessary updates. To do so, we compare the previous  
    // simulation State to the latest simulation data. If the two are different,  
    // update the UI element in the HUD View.  
    if (State.ScoreRed != scoreRed)  
    {  
        Format(text, "{0}", scoreRed);  
        ScoreRedMC.SetText(text); // Update the red team score text.  
  
        info.SetScale((scoreRed / (Double)MaxScore) * 100, 100);  
        TeamRedMC.SetDisplayInfo(info); // Update and scale the red team score  
                                         bar movieClip.  
    }  
  
    if (State.ScoreBlue != scoreBlue)  
    {  
        Format(text, "{0}", scoreBlue);  
        ScoreBlueMC.SetText(text); // Update the blue team score text.  
  
        info.SetScale((scoreBlue / (Double)MaxScore) * 100, 100);  
        TeamBlueMC.SetDisplayInfo(info); // Update and scale the blue team  
                                         score bar movieClip.  
    }  
}
```

```

}

if (State.ScoreRed != scoreRed || State.ScoreBlue != scoreBlue)
{
    // Update the "Winning" and "Losing" text fields for both teams
    // if the score has changed.
    if (scoreBlue > scoreRed)
    {
        RedWinningMC.SetText(LosingText);
        BlueWinningMC.SetText(WinningText);
    }
    else
    {
        RedWinningMC.SetText(WinningText);
        BlueWinningMC.SetText(LosingText);
    }
}

// Update the runtime clock
float secondsLeft = penv->GetSecondsLeftInRound();
unsigned sec = ((unsigned)secondsLeft % 60);
if (sec != State.Sec)
{
    unsigned min = ((unsigned)secondsLeft / 60);
    String timeLeft;
    Format(timeLeft, "{0}:{1:0.2}", min, sec);
    RoundTimeMC.SetText(timeLeft);
    State.Sec = sec;
}
}

```

```

Scaleform::String , Gfx::Value::SetText() scoreRed scoreBlue
textField . teamRed teamBlue X scale
Gfx::Value::SetDisplayInfo() . 0%
. 100%
, 100%
.

blueWinning textField redWinning textField
. HUDView "Winning" "Losing"
Gfx::Value::SetText() .

textField , UpdateView()
1 ,
Gfx::Value::SetText() roundTime textField String .

```

4.2 Player Stats Ammo

```
playerStats      HUDKit fla      Scene 1      playerStats      ,      textField
      (healthN, health),      (weapon), HUD      ,
      (reload)      .
```

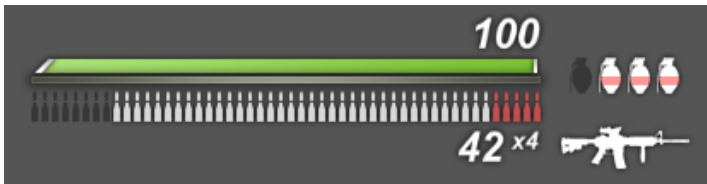


그림 6: playerStats 심볼

```
      MovieClips(clipN, ammoN, ammo, ammoBG, grenade,grenadeBG)
      . textFields clipN      ammoN
      .
```

```
ammo/healthVehicle      (ammo, ammoBG)      ,
```

```
      .      51      Keyframe      ,      Keyframe
Keyframe      . Gfx::Value::GotoAndStop()
MovieClip      Keyframe
      .
```

- *ammo*
- *ammoBG*

```
The grenade/armor      ammo      (grenade, grenadeBG)
      ,
      .      5      Keyframe      . Frame 1
      .      4      , Frame 5      . Gfx::Value::GotoAndStop\(\)
MovieClip      Keyframe
      .
```

- *grenade* displays the number of grenades remaining.
- *grenadeBG*

```
playerStats
```

```

void FxHUDView::UpdatePlayerStats(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();

    // Load latest player info.
    unsigned ammoInClip = pplayer->GetNumAmmoInClip();
    unsigned ammoClips = pplayer->GetNumClips();
    unsigned clipSize = pplayer->GetMaxAmmoInClip();
    unsigned grenades = pplayer->GetNumGrenades();

    unsigned ammoFrames = 51;
    unsigned grenadeFrames = 5;

    String text;

    if(pplayer->GetHealth() != State.Health)
    {
        unsigned health = unsigned(pplayer->GetHealth() * 100);
        Format(text, "{0}", health);
        HealthNMC.SetText(text); // Update the health text field.

        Value::DisplayInfo info;
        info.SetScale(Double(health), 100);
        HealthMC.SetDisplayInfo(info); // Update and scale the health bar.
    }

    if(ammoInClip != State.AmmoInClip)
    {
        Format(text, "{0}", ammoInClip);
        AmmoNMC.SetText(text); // Update the ammo text field.

        // The ammo is divided into two parts: the white bullet icons (AmmoMC)
        // and their respective backgrounds (AmmoBGMC).

        // To update the number of bullets currently in the clip, we use
        // GotoAndStop with AmmoMC and select the frame with the proper number
        // of white bullet icons. There are 51 frames in AmmoMC (defined
above        // in ammoFrames)and the first frame displays every bullet icon.

        // To display X number of bullets, we subtract X from the total number
        // of frames in AmmoMC (51) and GotoAndStop on the result.
        AmmoMC.GotoAndStop(ammoFrames - ammoInClip);
    }

    if(ammoClips != State.AmmoClips)
    {
        Format(text, "x{0}", ammoClips);
        ClipNMC.SetText(text); // Update the remaining ammo clips text field.
    }

    if(grenades != State.Grenades)
    {
        // Grenades are setup similar to Ammo. The first frame for the
        // Grenades movieClip shows 4 white grenade icons. The second frame

```

```

        // shows 3.

        // To display X number of grenades, we subtract X from the total number
        // of frames in GrenadeMC (5) and GotoAndStop on the result.
        // Note: GrenadeMC actually contains 10 frames but 6-10 are unused by
        // this implementation.
        GrenadeMC.GotoAndStop(grenadeFrames - grenades);
    }

    if (pplayer->GetWeaponType() != State.weaponType)
    {
        // Change the weapon icon if the player has changed weapons.
        unsigned weaponFrame = GetWeaponFrame(pplayer->GetWeaponType());
        WeaponMC.GotoAndStop(weaponFrame);

        // Update the ammo background icons based on clipSize.
        if (clipSize != State.AmmoClipSize)
            AmmoBGMC.GotoAndStop(ammoFrames - clipSize);
    }

    // If the ammo in clip is less than 6 bullets, play the "Reload" indicator
    // movieClip.
    if (ammoInClip <= 5)
    {
        // ReloadMC will play until we GotoAndStop on Frame 1, so no need to
        // start the animation more than once.
        if (!bReloadMCVisible)
        {
            bReloadMCVisible = true;
            ReloadMC.GotoAndPlay("on"); // Will cycle back to "on" frame
            when it reaches the end and play again.
        }
    }

    // If the reload indicator is displayed but there are more than six bullets
    // hide it and stop the animation by playing frame 1 of the movieClip.
    else if (bReloadMCVisible)
    {
        ReloadMC.GotoAndStop("1"); // Frame 1 contains stop();
        bReloadMCVisible = false;
    }
}

```

```

ammo      Frame 1      50      , Frame 51      .
                                Keyframe      HUD
                                ammoMC.GotoAndStop(ammoFrames - ammoInClip)      .

ammoBG      ,      X      Keyframe
ammoBGMC.GotoAndStop()      .      X      .
                                50      , ammoBG.GotoAndStop(1)      Frame 1      50

```

. MovieClip FxHUDView::UpdateView()

4.3

The *flagMC* HUDKit fla Scene 1 flag ,
. flagMC *flag* (arrow)
 , CaptureState .



그림 7: flagMC 심볼

flagMC Classic Tween Flash
. flagMC 5 Keyframe 4
 . Hidden, Visible, Fade-In, FadeOut. GotoAndStop() Keyframes
 / (1, 5, "On", "Off",)
 FxHUDView::UpdateEvents() *flagMC*

```
// Flag capture indicators and reticule behavior.
void FxHUDView::UpdateEvents(FxHUDEnvironment *penv)
{
    Value::DisplayInfo info;

    if (penv->IsHUDPlayerCapturingObjective())
    {
        if (!bFlagMCVisible)
        {
            SetVisible(&FlagMC, true); // Set the Flag MovieClip's
                                       // visibility to true.
            FlagMC.GotoAndPlay("on"); // Play the fade-in animation once.
            bFlagMCVisible = true;
        }

        // Shift the x-coordinate of the Flag Arrow to show the capture state
        // of the flag the player is currently capturing.
        info.SetX(penv->GetHUDCaptureObjectiveState() * 177);
        FlagArrowMC.SetDisplayInfo(info);
    }
}
```

```

        info.Clear();
    }
    else if (bFlagMCVisible & !penv->IsHUDPlayerCapturingObjective())
    {
        // If the flag indicator is still visible and the player is
        // no longer capturing an objective, play the fade-out animation
        FlagMC.GotoAndPlay("off");
        bFlagMCVisible = false;
    }
}

```

```

flagMC.GotoAndPlay("on")          flagMC          , Visible
    . flagMC.GotoAndPlay("off")          flagMC    11~20
    (          )          .          Frame 20          Frame
1          .          Flash          , ActionScript    C++
    . Frame 1          stop(); Frame 1          MovieClip
    .          Alpha    0          Frame 1          .

arrow    Gfx::Value::SetDisplayInfo\(\)          .
CaptureState          x          Gfx::Value::DisplayInfo          .

```

4.4

4.5

dirHit HUDKit.fla Scene 1 *dirHit* ,
 . *dirHit* 8 MovieClip
 . *tl, l, bl, b, br, r, tr, t.*

MovieClip ,

a)

b)

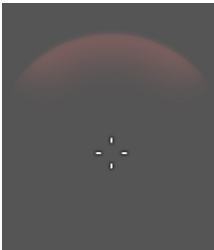


그림 9: *dirHit* 심볼

Flash 1
 Classic Tween .
 MovieClip Gfx::Value::GotoAndPlay("on") 1
 . MovieClip .

```
void FxHUDView::OnEvent( FxHUDEvent* pevent )
{
    switch ( pevent->GetType() )
    {
        // If the Damage Event (Player was hit) is fired, show the appropriate
        // directional hit indicator.
        // We can use GotoAndPlay("on") to play the fade-in animation. It
will
        // fade out on its own after ~1 second (this animation is setup on the
        // Flash timeline)
        case FxHUDEvent::EVT_Damage:
        {
            FxHUDDamageEvent* peventDamage = (FxHUDDamageEvent*)pevent;
            float dir = peventDamage->GetDirection();
            if (dir > 0)
            {
                if (dir > 90)
                {
                    if (dir > 135)
```

```

        DirMC_B.GotoAndPlay("on");
    else
        DirMC_BR.GotoAndPlay("on");
    }
else
{
    if (dir > 45)
        DirMC_TR.GotoAndPlay("on");
    else
        DirMC_T.GotoAndPlay("on");
    }
}
else
{
    if (dir < -90)
    {
        if (dir < -135)
            DirMC_B.GotoAndPlay("on");
        else
            DirMC_BL.GotoAndPlay("on");
    }
    else
    {
        if (dir < -45)
            DirMC_L.GotoAndPlay("on");
        else
            DirMC_TL.GotoAndPlay("on");
    }
    }
break;
}
}
}

```

4.6

```

expBar      HUDKit.fla    Scene 1      expBar      ,
                                                    . expBar      exp MovieClip
,
.

rank      HUDKit.fla    Scene 1      rank      ,
. rank      18      Keyframe      ,
Keyframe

```



그림 10: expBar 및 rank 심볼

```

void FxHUDView::UpdatePlayerEXP(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();
    if (pplayer->GetLevelXP() != State.Exp)
    {
        Value::DisplayInfo info;
        // Update the rank icon. Each rank icon is on a different frame of
        // the rank movieClip.
        RankMC.GotoAndStop(unsigned(pplayer->GetRank() + 1));
        info.SetScale(pplayer->GetLevelXP() * 100, 100); // Scale the EXP bar
                                                         movieClip.
        ExpBarMC.SetDisplayInfo(info);
    }
}

```

```

exp
.
Keyframe X      Gfx::Value::GotoAndStop()
Frame 0          X          + 1

```

4.7

```

The centerTextMC      HUDKit.fla    Scene 1      centerText      ,
textField
centerText

```

You were killed! Respawning...

그림 11: centerTextMC 심볼

```
centerTextMC      textField      centerText MovieClip      textField
MovieClip
    "on5"  "on"      Keyframe      , labeled
    5      , "on"      3
    .      centerTextMC      alpha  0      Frame 1
MovieClip      , "on"      "on5"      MovieClip
TextFieldMC.SetText(RespawnText); // Set the text field of the Center Text.
CenterTextMC.GotoAndPlay("on"); // Play the fade-in animation, it will fade on its
                                own after ~3 sec.
```

```
centerTextMC      textField      ,      MovieClip      GotoAndPlay("on")
    .      Classic Tween      MovieClip      3
```

4.8

```
popup      HUDKit.fla      Scene 1      popup      ,
    .      3      (      )
popup      .      , C++
3~5      .
```



그림 12: popup 심볼

```
Popup      .      Classic
Tweens      popup      . popup      actions, popupNumber,
popupText,popupBG 4      . popupBG      "      "
    . popupText      popupNumber
```

textField

textFieldds

GotoAndPlay()

"on"

Keyframe

Frame 1

Alpha 0

```
// If a KillStreak event is fired, display the kill streak pop-up.  
case FxHUDEvent::EVT_KillStreak:  
{  
    FxHUDKillStreakEvent* peventKS = (FxHUDKillStreakEvent*)pevent;  
  
    // Format the number of kills  
    String text;  
    Format(text, "{0}", peventKS->GetSrcKillStreak());  
    PTextFieldMC.SetText(text); // Set the text field of the PopUp.  
  
    // Play the fade-in animation, it will fade on its own after ~4 sec.  
    PopupMC.GotoAndPlay("on");  
}  
break;
```

FxHUDEvent KillStreak

cast

popupText *textField*

PTextFieldMC

popup

GotoAndPlay("on") C++

4.9

log

HUDKit.fla

Scene 1

log

, HUD

3



그림 13: log 및 logMessage 심볼

```

        logMessage
        3
        . C++
        ,
htmlText        textField

        FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip()
LogMessage MovieClip        Log MovieClip

        .

FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip()
{
    if (MessageMCs.IsEmpty())
    {
        // Request a new line in the SWF
        Value logline;
        LogMC.AttachMovie(&logline, "LogMessage", "logMessage"+NumMessages);
        FxHUDMessage::Movieclip* pmc = new FxHUDMessage::Movieclip(logline);
        MessageMCs.PushBack(pmc); // Add new Message MC to list of unused
                                   message movieClips.
    }
    FxHUDMessage::Movieclip* pmc = MessageMCs.GetFirst(); // Use an unused Message
                                                            movieClip.
    pmc->SetVisible(true);
    MessageMCs.Remove(pmc);
    return pmc;
}

        C++
        .        Level Up
        "You are now a [Rank Icon] [Rank Name]"
        C++
        .
void    FxHUDLog::Process(FxHUDEvent *pevent)
{
    String msg;
    switch (pevent->GetType())
    {
        case FxHUDEvent::EVT_LevelUp:
        {
            FxHUDLevelUpEvent* e = (FxHUDLevelUpEvent*)pevent;
            Format(msg, "You are now a <img src='rank{1}'/> {0}!",
                e->GetNewRankName(), e->GetNewRank());
        }
        break;
    }
}

FxHUDMessage::Movieclip* pmc = GetUnusedMessageMovieclip();
FxHUDMessage* m = new FxHUDMessage(msg, pmc); // 3 seconds

```

```
Log.PushBack(m);
NumMessages++;
```

HUD Update

```
void    FxHUDLog::Update()
{
    SF_ASSERT(LogMC.IsDisplayObject());

    Double rowHeight = 30.f;

    // Tick the message lifetimes
    Double yoff = 0;
    FxHUDMessage* data = Log.GetFirst();
    while (!Log.IsNull(data))
    {
        FxHUDMessage* next = Log.GetNext(data);
        if (data->IsAlive())
        {
            // Layout mgmt and animation
            Value::DisplayInfo info;
            info.SetY(yoff);
            data->GetMovieclip()->GetRef().SetDisplayInfo(info);
        }
        data = next;
        yoff += rowHeight;
    }

    // Layout management and animation
    Value::DisplayInfo info;
    info.SetY(LogOriginalY - (NumMessages * rowHeight));
    LogMC.SetDisplayInfo(info);

    // Remove dead messages
    data = Log.GetFirst();
    while (!Log.IsNull(data))
    {
        FxHUDMessage* next = Log.GetNext(data);
        if (!data->IsAlive())
        {
            Log.Remove(data);
            NumMessages--;

            FxHUDMessage::Movieclip* pmc = data->GetMovieclip();
            pmc->SetVisible(false);
            MessageMCs.PushBack(pmc);

            delete data;
        }
        data = next;
    }
}
```



```

    Value billboard, temp;
    String instanceName;
    Format(instanceName, "{0}{1}", BillboardPrefix, BillboardCount++);
    CanvasMC.AttachMovie(&billboard, SymbolLinkage, instanceName);

    BillboardMC* pbb = new BillboardMC();
    pbb->Billboard = billboard;
    billboard.GetMember("label", &temp);
    temp.GetMember("textField", &temp);
    pbb->Textfield = temp;

    pbb->Billboard.SetDisplayInfo(info);
    UnusedBillboards.PushBack(pbb);
}

```

```

    BillboardCache    CanvasMC    MovieClip
    .
    billboard_enemy/billboard_friendly    MovieClip    CanvasMC.AttachMovie()
    .

    . UpdateBillboards(), BeginProcessing(),
EndProcessing()    GetUnusedBillboardMovieclip(). BeginProcessing()    EndProcessing()
    .

```

```

    UpdateBillboards()
    .

// Process friendlies
// Friendly billboards will always show names

BillboardMC* pbb = NULL;

FriendlyBillboards.BeginProcessing();
for (unsigned i=0; i < friendlies.GetSize(); i++)
{
    info.Clear();
    if (FriendlyBillboards.GetUnusedBillboardMovieclip(friendlies[i].pPlayer,
                                                         &pbb))
    {
        info.SetVisible(true);
        pbb->Billboard.GotoAndPlay("show");
        // Load player info
        String title;
        Format(title,
               "<img src='rank{0}' width='20' height='20' align='baseline'
               vspace='-7' /> {1}",
               friendlies[i].pPlayer->GetRank()+1,
               friendlies[i].pPlayer->GetName());
    }
}

```

```

        pbb->Textfield.SetTextHTML(title);
    }
    info.SetX(friendlies[i].X);
    info.SetY(friendlies[i].Y);
    info.SetScale(friendlies[i].Scale, friendlies[i].Scale);
    pbb->Billboard.SetDisplayInfo(info);
}

FriendlyBillboards.EndProcessing();

UpdateBillboards()
    .
    . GotoAndPlay("show") , MovieClip X, Y
    . Gfx::Value::SetDisplayInfo() , Gfx::Value::SetText()
    .

textField
    . htmlText
    .
    . src='rank{0}' HTML
    .
    . friendlies[i].pPlayer->GetRank()+1 +1 . img
    . , , vspace . {1}
    . friendlies[i].pPlayer->GetName() .

```

5 미니맵 뷰

C++ (Apps\Kits\HUD\)

- **FxMinimap.h:**
- **FxMinimapView.h/.cpp:**

5.1.1 미니맵 뷰

- **FxMinimapIconType:** ID
- **FxMinimapIcon:**
- **FxMinimapIconCache:**

```

/
MinimapIconFactoryBase
MovieClip
MovieClip
MovieClip
, textField
Update()

```

- o *PlayerIcon:*
- o *FlagIcon*
- o *ObjectiveIcon*
- o *WaypointIcon*

- **FxMinimapView:**
 - UpdateView()
 - PlayerIcon::Update() (FxMinimapView.cpp; 116). Direct Access
- API . *MovieClip*
- MovieClip . SetDisplayInfo

```

SetMember()                                MovieClip                                (
,      )                                .      Gfx::Movie::SetVariable      Gfx::Value::SetMember
.

```

```

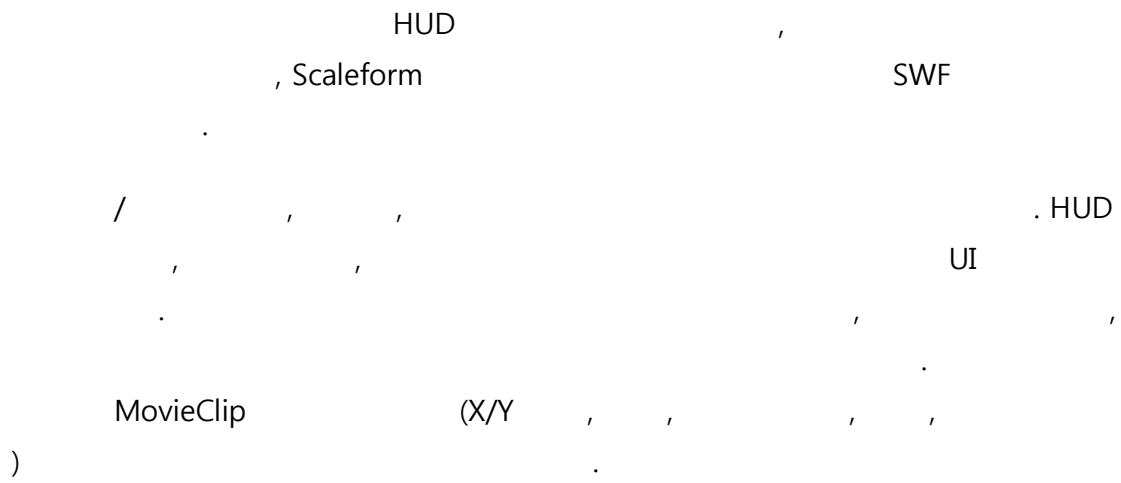
virtual void      Update(FxMinimapView* pview, FxMinimapEntity* pentity,
                        float distSquared)
{
    SF_UNUSED(distSquared);
    bool hasIcon = (pentity->GetIconRef() != NULL);
    PointF pt = pentity->GetPhysicalPosition();
    bool showDir = pview->IsShowingPlayerDir();

    // If entity did not have an icon attached before, then wait for the
    // next update to set the state data. picon is most probably not initialized
    // (the movie needs to advance after creation)
    if (hasIcon && (State != (int)showDir))
    {
        // state 0: no arrow, state 1: show arrow
        Value frame(Double(showDir ? 2.0 : 1.0));
        MovieClip.Invoke("gotoAndStop", NULL, &frame, 1);
        State = (int)showDir;
    }

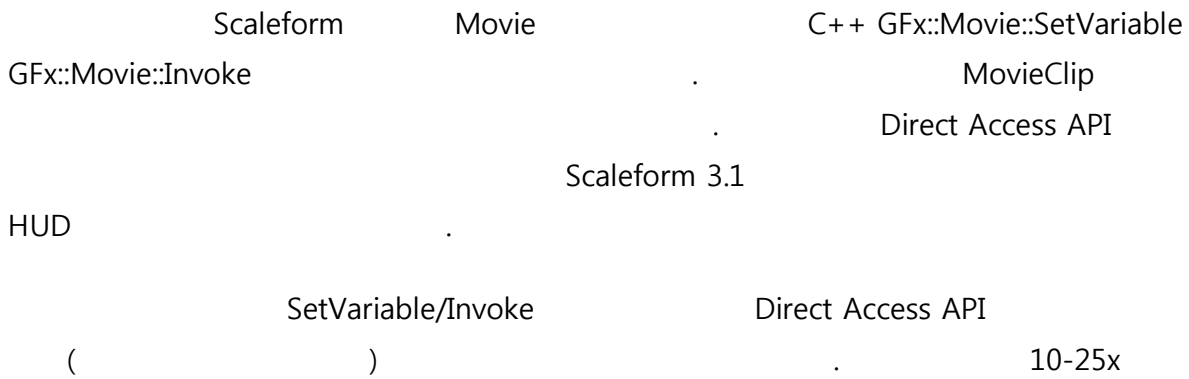
    pt = pview->GetIconTransform().Transform(pt);
    Value::DisplayInfo info;
    if (State)
    {
        info.SetRotation(pview->IsCompassLocked() ?
                        (pentity->GetDirection() + 90) : (pentity->GetDirection() -
                        pview->GetPlayerDirection()));
    }
    info.SetPosition(pt.x, pt.y);
    MovieClip.SetDisplayInfo(info);
}

```

6 성능 분석



6.1



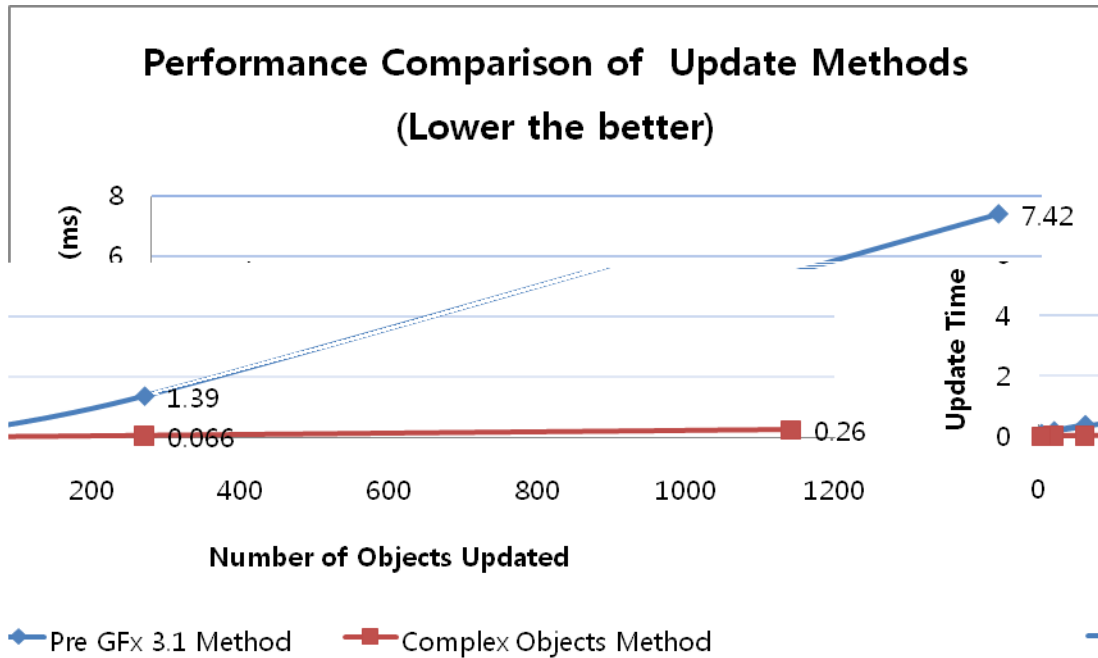
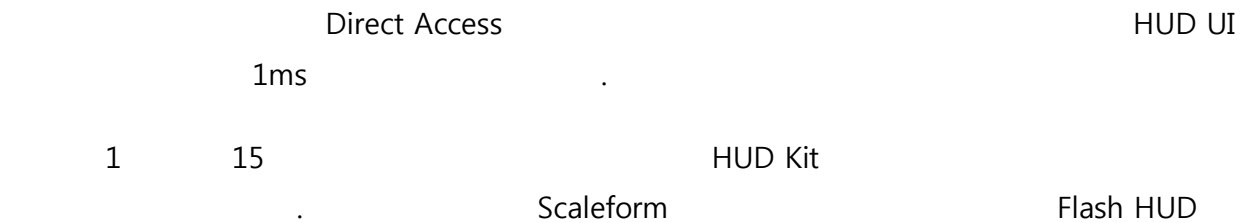


그림 15: 미니맵에 대한 Direct Access API 와 SetVariable/Invoke 업데이트 메소드 성능 비교



플랫폼	FPS	업데이트(ms)	디스플레이(m s)	Advance(ms)	총합(ms)
Windows Vista	1148	0.018	0.512	0.017	0.527
MacBook Pro					
Xbox 360	1136	0.032	0.454	0.010	0.497
PlayStation 3	752	0.044	0.668	0.021	0.733

표 1: Scaleform HUD Kit 플랫폼별 평균 성능 통계



6.2 Memory Breakdown

HUD	Memory Breakdown	.
GfxExport	(Export)	.GFX
(Export)	.	,

1. 파일 형식: GFX 기본 보내기, 압축되지 않은 이미지

- : 1858kb
- HUDKit.gfx - 33kb
- Minimap.gfx - 50kb
- fonts_en.gfx - 46kb
- gfxfontlib.gfx - 100kb
- HUDKit.gfx
 - o UI - 603kb
 - o - 72kb
 - o - 51kb
 - 참고: HUD (43kb)
- Minimap.gfx
 - o - 644kb
- Map.jpg - 259kb